Lab 03 -- D3.js: Scales, Axes, and Visual Encoding

International University

Data Science and Data Visualization

Lab 03 — D3.js: Scales, Axes, and Visual Encoding

Work mode: Groups of 2
In-class duration: 2.5 hours (150 minutes)
Homework: 1 week, up to 3 hours per group
Points: 100
Main topic: Scales, axes, numeric data, proportional symbols, categorical color, and verification

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1. Learning Objectives

By the end of this lab, you should be able to:

LO1 — Explain scales

Explain the difference between a scale's **domain** (input values) and **range** (output values), and predict how a change to either affects a chart.

LO2 — Build a scalable scatterplot

Use D3 linear scales to map population to x-position and GRDP per person to y-position, so the chart remains useful when data values change.

LO3 — Use axes meaningfully

Create labeled x- and y-axes with readable ticks, correct orientation, and suitable spacing.

LO4 — Encode additional variables

Use a square-root scale so circle area represents province area, and a discrete color scale so color represents population density.

LO5 — Verify, debug, and adapt

Test the chart with different datasets, identify misleading or broken encodings, and modify the chart when a requirement changes.

LO6 — Explain and own the solution

Explain the key scales and visual encodings and complete a small individual change during the Group Check.

> **Main learning bottleneck:** Can you explain why each province appears at that position, with that circle size and color—and whether those visual choices are truthful?

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2. Lab Flow

```
THINK → BUILD → VERIFY → ADAPT → GROUP CHECK
 AI OFF     AI ON    AI ON    AI ON    OWNERSHIP
```

Phase	Time	AI
THINK	25 min	OFF
BUILD	60 min	ON
VERIFY	25 min	ON
ADAPT	20 min	ON
GROUP CHECK	15 min	Instructor-defined
Buffer / transition	5 min	—
Total	**150 min**	

Working principle: Think first. Ask AI second. Verify always. Own the result.

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3. Work in Groups of 2

One student is the **Driver** (writes and runs code); the other is the **Navigator / Reviewer** (checks assumptions, asks why, and proposes tests). Switch roles at least once.

Both students must be able to explain the implementation. Pair work does not mean one person codes while the other watches.

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4. AI Policy

THINK — AI OFF

Do not use ChatGPT, Claude, Copilot, or other AI coding assistants. Discuss with your partner, use your own notes, and ask the instructor to clarify the requirements.

BUILD / VERIFY / ADAPT — AI ON

AI may explain D3 syntax, suggest an implementation, review code, or help debug. You must inspect and test any suggested code. Do not submit code that neither partner can explain.

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5. Scenario

A data team wants to explore differences between provinces in Viet Nam. Build a scatterplot in which each point represents one province:

- **x-position:** population;
- **y-position:** GRDP per person (million VND/person/year, as specified by the source dataset);
- **circle area:** province area;
- **circle color:** population density, using discrete color categories.

The chart must use the source CSV:

`https://tungth.github.io/data/vn-provinces-data.csv`

The original lab specifies an SVG canvas of **300 pixels wide × 600 pixels high**. Keep this size for the initial implementation, but make the chart dimensions easy to change for the adaptation task.

Important: inspect the CSV headers and values before coding. Use the actual field names and units in the downloaded dataset; do not guess column names. The original handout uses "GRPD" in the heading, but the intended measure is **GRDP** (Gross Regional Domestic Product).

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6. Setup — 10 Minutes

Create this project:

```
lab03-d3/
 index.html
 script.js
 style.css
```

Use D3 for the visualization. Do not use another charting library.

References:

- D3: https://d3js.org/
- D3 tutorials: https://github.com/d3/d3/wiki/Tutorials
- D3 scales: https://www.d3indepth.com/scales/
- D3 scale and color: https://github.com/d3/d3-scale-chromatic

Read the assigned material from *Interactive Data Visualization for the Web*, Chapters 7–8 (Scales and Axes).

Run the page using a local web server. If CSV loading fails, check the browser console and Network tab before changing the code.

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7. THINK — AI OFF — 25 Minutes

Write short answers in your own words before opening an AI assistant.

7.1 Domain and range

Consider:

```
const xScale = d3.scaleLinear()
  .domain([0, 1000])
  .range([40, 560]);
```

1. What is the input domain?
2. What is the output range?
3. What output should `xScale(0)` produce?
4. What output should `xScale(1000)` produce?
5. Approximately what output should `xScale(500)` produce?
6. If the population maximum changes from 1,000 to 2,000, why might a dynamic domain be better than keeping `[0, 1000]`?

7.2 Why invert the y-scale?

Consider:

```
const yScale = d3.scaleLinear()
  .domain([0, 100])
  .range([260, 20]);
```

1. Why does the range go from `260` to `20` rather than `20` to `260`?
2. Where will a value of `100` appear: near the top or near the bottom?
3. What misleading result could occur if the y-scale is not inverted?

7.3 Choose the correct visual encoding

Match each data field to its intended visual property:

Data field	Visual property
Population	x-position
GRDP per person	y-position
Province area	Circle area
Population density	Discrete color

Then answer:

1. Why is circle area a more appropriate description than "circle radius is proportional to area"?
2. Why should the color scale be discrete if the task asks for categories?
3. What must you inspect in the CSV before defining the scales?

7.4 Predict before running

If the CSV contains 63 valid province records and the code uses one circle per record, how many circles should appear? What might cause fewer circles to appear?

Keep your answers. The instructor may ask you to explain one during the Group Check.

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8. BUILD — AI ON — 60 Minutes

8.1 Create the page and load the data

The page must include:

- page title: **Lab 3**;
- your name and student ID;
- course title: **Data Science and Data Visualization**;
- lab title: **Lab 3 — D3.js: Scales, Axes, and Visual Encoding**;
- the statement: **"This is all my own work. I did not copy the code from any other source."**

Load the CSV with D3. Inspect the header names and at least two records in the console. Convert numeric fields from strings to numbers before using them in scales. Handle loading errors visibly or in the console.

8.2 Build the scatterplot

Your chart must satisfy all of the following:

- ☐ Load `vn-provinces-data.csv` using D3.
- ☐ Use an SVG initially sized 300 pixels wide × 600 pixels high, as specified in the original handout.
- ☐ Draw one circle for each valid province record.
- ☐ Map population to the x-position with a linear scale.
- ☐ Map GRDP per person to the y-position with a linear scale; larger values must appear higher on the chart.
- ☐ Derive scale domains from the actual data rather than hard-coding observed minimum and maximum values.
- ☐ Leave enough inner space for axes and labels (use margins/padding).
- ☐ Draw a bottom x-axis and a left y-axis.
- ☐ Add readable axis titles and tick labels with units where appropriate.
- ☐ Map province area to circle **area** using a square-root scale or an equivalent mathematically justified mapping.
- ☐ Map population density to discrete colors using a documented set of categories or thresholds.
- ☐ Include a legend that explains the density colors.
- ☐ Include a chart title and a short note describing the source and units.

8.3 Recommended implementation structure

Organize the code so another developer can find these parts:

```
load CSV↓

inspect / convert data types↓

set SVG dimensions and margins↓

create xScale, yScale, areaScale, densityColor↓

draw circles↓

draw axes and axis titles↓

draw legend and source note
```

You may use a different structure if it remains readable and all requirements are met.

8.4 Important design decisions

- **Dynamic domains:** use the dataset to calculate the domain. Decide whether to include zero based on the meaning of the measure and explain your choice.
- **Y-axis direction:** SVG y-coordinates increase downward, so the output range normally needs to be reversed for a conventional scatterplot.
- **Circle size:** if the encoded quantity is area, a square-root scale is appropriate because circle area is proportional to radius squared. Do not map area linearly to radius without justification.
- **Density colors:** use clear, distinguishable categories and show their meaning in a legend. Define category thresholds from the data or a clearly stated rule; do not silently invent thresholds.
- **Overplotting:** if circles overlap, consider opacity, a thin stroke, or an explanation of the limitation. Do not remove records simply to make the chart look cleaner.

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9. VERIFY — AI ON — 25 Minutes

Do not ask AI "Is my code correct?" as your only test. State a hypothesis, run a test, and record the evidence.

9.1 Verification checklist

☐ The number of circles equals the number of valid records loaded.
☐ The browser console shows no uncaught errors.
☐ Population and GRDP fields are numeric, not strings.
☐ The smallest and largest values fall within the plotting area.
☐ The y-axis increases in value from bottom to top.
☐ The axes align with the chart's plotting area.
☐ Every density category in the legend matches the circle colors.
☐ A larger province area produces a larger circle area, not merely a proportionally larger radius.
☐ Axis titles and units are understandable without reading the code.

9.2 Test with small data

Temporarily replace the loaded data with a small test dataset using the same field structure as your code. Include at least three records with:

- a low population and low GRDP value;
- a medium population and medium GRDP value;
- a high population and high GRDP value.

Also test one record with the maximum area and one record with a density exactly on a category boundary. Confirm that the positions, sizes, and colors behave as expected. Restore the CSV before submission.

9.3 Debugging challenge

Your instructor will assign one defect, or you may choose one below:

Bug A — Fixed domain: the chart uses `[0, 1000]` for population even though some values exceed 1,000.

Bug B — Reversed y-axis: larger GRDP values appear lower on the chart.

Bug C — Misleading circles: the code maps province area directly to circle radius.

Bug D — Incorrect types: CSV values are still strings when used in numeric calculations.

Bug E — Missing records: the number of circles is lower than the number of valid rows.

Bug F — Unclear colors: the chart uses multiple colors but has no legend or defined category rule.

For your assigned bug, record:

1. What did you observe?
2. What did you expect?
3. What is the likely cause?
4. What change did you make?
5. How did you verify the fix?

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10. ADAPT — AI ON — 20 Minutes

The data team has changed its request. Make the following changes without rewriting the whole chart.

Required adaptation

Change the chart size from 300 × 600 to 450 × 900 pixels.

Update the dimensions and scale ranges so that the chart remains correctly laid out. The axes, circles, and labels must stay within the plotting area. The data-to-visual relationships must remain correct.

Choose one additional adaptation

Choose **one**:

- **A. Highlight:** visually highlight the province with the highest GRDP per person and explain how you identified it.
- **B. Tooltip:** show the province name, population, GRDP per person, area, and density when a user hovers over a circle.
- **C. Dynamic color thresholds:** change the density categories and update both the color mapping and legend consistently.

Before using AI, write one sentence predicting which parts of the code should change. After implementing the change, verify it with at least one test.

Reflection

1. Which part was easiest to adapt?
2. Which part depended on hard-coded assumptions?
3. What would you improve if the dataset gained more records or much larger values?

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11. GROUP CHECK — 15 Minutes

The instructor may ask either partner to explain or modify part of the solution. You may not rely on your partner to answer every question.

Possible prompts:

- Explain the domain and range of your x-scale.
- Why is the y-scale range reversed?
- Explain how one CSV row becomes one circle.
- Why did you choose a linear scale for population?
- Why use a square-root scale for circle size?
- How are density categories defined and represented in the legend?
- Show how you verified that all records are represented.
- What would break if the data contained a much larger population value?
- Make one small change to a scale or axis and explain the result.

Ownership standard: explain the reason for a design decision, identify one limitation, and demonstrate one small change.

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12. Submission

Submit the following to the Blackboard assignment:

- ☐ all HTML, JavaScript, and CSS files;
- ☐ the completed chart with axes, labels, size encoding, and density legend;
- ☐ a short `README.md` or text section listing your scale choices and density-category rule;
- ☐ a brief verification note containing the bug tested, the fix, and the evidence;
- ☐ the names and student IDs of both group members.

Do not submit only a screenshot. The instructor must be able to run and inspect your work.

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13. Assessment — 100 Points

Component	Points	Evidence
THINK	15	Individual predictions and scale reasoning
BUILD	30	Correct data loading, scales, axes, size/color encoding
VERIFY	20	Tests, debugging evidence, correct data representation
ADAPT	15	Responsive chart-size change and one extra adaptation
GROUP CHECK	20	Individual explanation and live modification
Total	**100**	

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14. Homework — Up to 3 Hours per Group

Extend the visualization with one focused feature:

- add a search or filter that highlights provinces matching a user-selected density category; **or**
- add a tooltip and a small written interpretation of two interesting data points.

Requirements:

1. Keep the original encodings correct.
2. Test at least three cases, including an edge case.

3. Explain one limitation of the visualization (for example, overlap, area perception, or differences in units).
4. Include a short note describing where AI helped and what your group independently verified.

Do not add a new framework, database, authentication, or deployment requirement. The goal is to deepen understanding of D3 scales and axes, not to build a larger application.